Text 1
Fraud Detection in Transaction Graphs using GNNs and Graph Heuristics

Text 3
Project Description

This project focuses on detecting fraudulent behavior in transaction networks by combining classical graph algorithms with Graph Neural Networks (GNNs). Transactions are modeled as a graph, where nodes represent accounts (or transactions) and edges represent interactions or fund transfers.

Text 4
To capture structural patterns in the network, graph-based features such as degree, clustering coefficient, and PageRank are computed using classical algorithms. These features help identify suspicious behavior and are used both for heuristic-based scoring and as input to a GNN.

Text 5
The GNN learns representations by leveraging both node features and graph structure, enabling it to detect complex relational patterns that traditional methods may miss. The final system combines predictions from heuristic methods and the GNN to classify nodes as fraudulent or legitimate.

Text 6
Objective

To build a robust fraud detection system that effectively integrates algorithmic graph insights with machine learning, demonstrating the advantage of combining data structures & algorithms with modern deep learning techniques.

Text 7
1 Dataset Description

1.1 Overview - Elliptic Bitcoin Transaction Dataset

The dataset used in this project is the **Elliptic Bitcoin Transaction Dataset**, a

Text 8
real-world graph dataset of Bitcoin transactions labeled by the blockchain analytics company *Elliptic*. We have taken it from kaggle.

Text 9
The dataset is provided in three CSV files:

Text 10
- **elliptic_txs_classes.csv**
  Contains labels for each transaction:

Text 11
  - 1 → Illicit (Fraud)
  - 2 → Licit (Not fraud)
  - Unknown → Unlabeled
- **elliptic_txs_features.csv**
  Contains feature information for each transaction node, including 165 anonymized features and a time step index.
- **elliptic_txs_edgelist.csv**
  Contains directed edges representing fund flow between transactions:

Text 12
$$txId_1 \rightarrow txId_2$$

Text 13
1.2 Exploratory Data Analysis (EDA)

Text 14
Exploratory Data Analysis was performed in a (`eda_for_data.py`) file to understand the dataset structure, distribution, and graph properties.

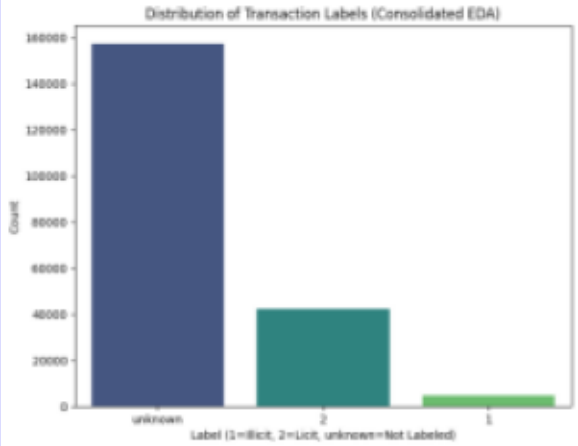
Text 15
1.2.1 Classes File

Text 16
- Total transactions: **203,769**

Text 17
Class distribution:

Text 18
- Unknown: 157,205
- Licit: 42,019
- Illicit: 4,545

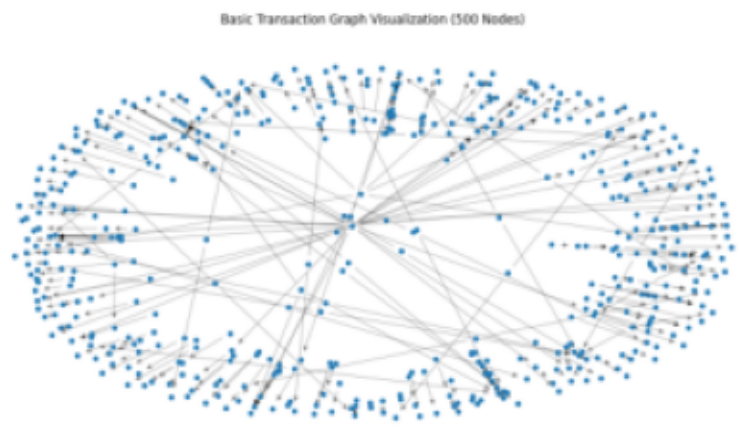
Image 1



Text 1

Figure 1: Distribution of transaction lab

Image 3



Text 10

Figure 3: Distribution of transactions across timestamps

Text 2

1.2.2 Features File

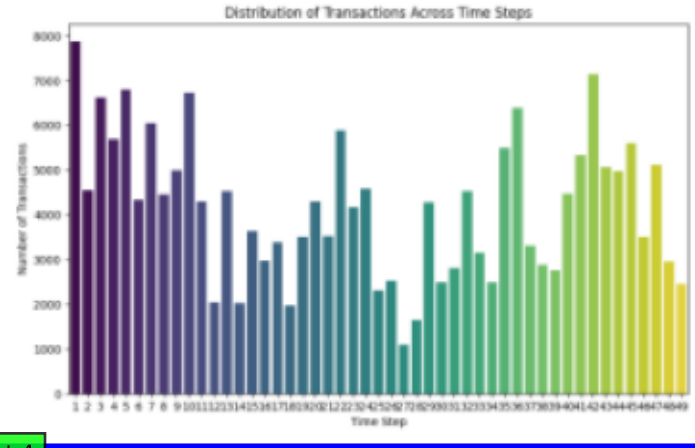
Text 3

- Dataset shape: **(203,769 × 167)**
- Features per transaction: **165**

Each row contains:

- Transaction ID (txId)
- Time step
- 165 anonymized features

Image 2



Text 4

Figure 2: Distribution of transactions across timestamps

Text 5

1.2.3 Edge List File

Text 6

Represents directed transaction relationships between nodes:

Text 7

$$txId_1 \rightarrow txId_2$$

Text 8

Using the NetworkX library:

- Nodes represent transactions
- Edges represent fund flows

A sample graph was constructed and visualized with:

Text 9

- 500 nodes
- 367 edges

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2 Graph Heuristics — Feature Engineering via Graph Algorithms

Text 12

A key contribution of this project is the extraction of structural graph features using classical graph algorithms. These features capture relational and behavioral patterns that are not visible through node attributes alone. They are used both as standalone inputs for machine learning models and as additional features for the Graph Neural Network (GNN).

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All heuristics are computed on the directed transaction graph

Text 14

$$G = (V, E)$$

Text 15

where V represents transactions and E represents directed fund flows.

Text 16

2.1 Degree Features (In-Degree & Out-Degree)

Out-degree: Number of outgoing edges from a node

Text 17

In-degree: Number of incoming edges to a node

Text 18

These features quantify how funds move through a transaction.

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- High out-degree → fan-out behavior (money dispersion)
- High in-degree → aggregation of funds

Both features are computed using adjacency list traversal in

Text 1
$O(V + E)$

Text 2
time.

Text 3
2.2 Strongly Connected Components (SCC — Kosaraju's Algorithm)

Strongly Connected Components identify cycles in the transaction graph.

Text 4
In blockchain systems, legitimate transactions typically form acyclic flows, while cycles often indicate suspicious activity such as fund recycling or wash trading.

Text 5
Two features are derived:

- **SCC Size Score:** Normalized size of the component
- **SCC Density:** Internal connectivity of the component

These features are computed using Kosaraju's algorithm with complexity

Text 6
$O(V + E)$

Text 7
2.3 Topological Depth (Kahn's Algorithm)

Topological depth measures the position of a transaction within the flow of funds.

Text 8
- Depth $= 0 \rightarrow$ source transactions
- Higher depth $\rightarrow$ downstream or relay transactions

Observed fraud patterns:

- Moderate depth $\rightarrow$ intermediary relay nodes
- Very high depth $\rightarrow$ long laundering chains

Nodes belonging to cycles (non-trivial SCCs) are assigned

Text 9
$depth = -1$

Text 10
to explicitly mark structural anomalies.

This feature is computed using a BFS-based Kahn's algorithm.

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2.4 PageRank (Graph Centrality)

PageRank assigns an importance score based on incoming connections and their quality.

Text 12
Text 13
- High PageRank $\rightarrow$ transaction acts as a hub
- Indicates central routing points in fund flow

This feature is particularly useful for identifying nodes involved in layering stages of money laundering.

Text 14
PageRank is computed iteratively using:

Text 15
- Damping factor:

Text 16
$d = 0.85$

Text 17
- Convergence threshold:

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10^{-4}

Text 19
Time complexity:

Text 20
$O(iterations \times (V + E))$

Text 21
2.5 Clustering Coefficient

The clustering coefficient measures how tightly connected a node's neighborhood is.

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$$CC = \frac{2 \times number of links between neighbors}{k(k-1)}$$

Text 23

Text 24
where k represents the number of neighboring nodes.

Text 25
- High value $\rightarrow$ node belongs to a dense cluster
- Indicates coordinated or group-based fraudulent activity

This feature is computed on the undirected neighborhood (combined in/out neighbors).

Text 26
Nodes with fewer than two neighbors are assigned

Text 27
$CC = 0$

Text 28
2.6 Motif-Based Features

Motifs are small recurring structural patterns that act as behavioral signatures within transaction networks.

Text 29
The following motif-based features were extracted:

Text 30
- **Chain Score:**

Text 1
$in - degree \times out - degree$

Text 2
Captures relay behavior (pass-through transactions)

- **In-Star Score:**

Text 3
$in - degree2$

Text 4
High for nodes aggregating funds from multiple sources

- **Out-Star Score:**

Text 5
$out - degree2$

Text 6
High for nodes distributing funds widely

- **Triangle Score**
  Counts closed triplets indicating cyclic routing patterns (fraud loops)

These features provide localized structural fingerprints of transaction behavior.

Text 7
2.7 BFS Layering Score

Text 8
A bounded Breadth-First Search (depth ≤ 5) is used to measure multi-path reachability.

Text 9
This feature tracks how often downstream nodes are reached through different traversal paths.

Text 10
- High score $\rightarrow$ dense overlapping connectivity
- Such patterns are uncommon in normal transaction graphs
- May indicate coordinated transactional structures

Scores are normalized to the range:

Text 11
$[0, 1]$

Text 12
2.8 DFS Flow Score

Text 13
A bounded Depth-First Search (depth ≤ 5) captures flow complexity and cyclic behavior.

Text 14
Scoring strategy:

- Visiting a new node $\rightarrow +1$
- Revisiting a node (cycle detected) $\rightarrow +5$

This weighted scoring makes the feature highly sensitive to:

Text 15
- Cycles
- Repeated traversal paths

Neighbor exploration is capped to prevent exponential growth during traversal.

Text 16
Table 1: Summary of Graph-Based Heuristic Features Used for Fraud Detection

Table 1

	Algorithm	Fraud Signal
out_degree	Adjacency traversal	Fund dispersion (fan-out)
in_degree	Adjacency traversal	Fund aggregation
pagerank	Iterative algorithm	Transaction hub centrality
depth	Kahn's algorithm	Position in transaction chain
clustering	Local clustering coefficient	Dense fraud group membership
chain_score	Motif-based feature	Relay behavior
in_star_score	Motif-based feature	Aggregation pattern
out_star_score	Motif-based feature	Dispersion pattern
triangle_score	Motif-based feature	Cyclic fund routing
dfs_flow_score	Bounded DFS	Cycle-rich neighborhood
bfs_layering_score	Bounded BFS	Multi-path connectivity
scc_size	Kosaraju's SCC	Cycle membership
scc_density	Kosaraju's SCC	Internal density of cycles

Text 17
3 Model Training and Evaluation — Random Forest on Graph Heuristics

3.1 Model Overview

To evaluate the effectiveness of graph-based heuristic features, a Random Forest Classifier was trained using the engineered features derived from the transaction graph.

Text 18
Random Forest was selected for the following reasons:

Text 19
- It handles non-linear relationships between features effectively
- It is robust to feature scaling issues
- It provides feature importance scores, enabling interpretability of graph-based heuristics

Text 20
3.2 Data Preparation

The dataset used for training the model was:

Text 21
`graph_heuristics_features.csv`

Text 22
Preprocessing steps performed:

- Unlabeled transactions (label = -1) were removed from the dataset
- Feature set consisted of the first **13 engineered heuristic features**
- Target variable:
  - $0 \rightarrow$ Legitimate transaction
  - $1 \rightarrow$ Fraudulent transaction

The dataset was split as follows:

- Training set: 75%
- Test set: 25%

3.3 Model Configuration

The Random Forest classifier was trained using the following hyperparameters:

- Number of trees: 300
- Maximum depth: 10
- Minimum samples required to split: 5
- Class weighting: balanced
  This helps address class imbalance in the fraud detection dataset.
- Random state: 42 Ensures reproducibility of results.

Instead of the default classification threshold of 0.5
a probability threshold of 0.6

The method was used to reduce false positive predictions and improve the reliability of fraud detection. We get the following confusion matrix for the random forest model.

3.4 Evaluation Metrics

We trained and tested with random forests at different depths and splits and got the best model with an accuracy of 83.4%.

Image 1

Accuracy: 0.8347221029121209

	precision	recall	f1-score	support
0	0.96	0.85	0.90	10479
1	0.34	0.68	0.45	1162
accuracy			0.83	11641
macro avg	0.65	0.76	0.68	11641
weighted avg	0.90	0.83	0.86	11641

Confusion Matrix:

	Pred 0	Pred 1
Actual 0	8931	1548
Actual 1	376	786

Figure 4: Xg Boost model accuracy

Based on the evaluation results obtained from the Random Forest classifier, the following observations were made:

- High precision was achieved for legitimate transactions, indicating that the model produces very few false positive predictions.
- Moderate recall was observed for fraudulent transactions, showing that the model successfully detects a significant portion of fraud cases.
- Precision for fraudulent transactions was comparatively lower, suggesting the presence of some false alarms during classification.
- This trade-off between precision and recall is expected due to the inherent class imbalance present in the dataset.

3.5 Confusion Matrix

Image 2

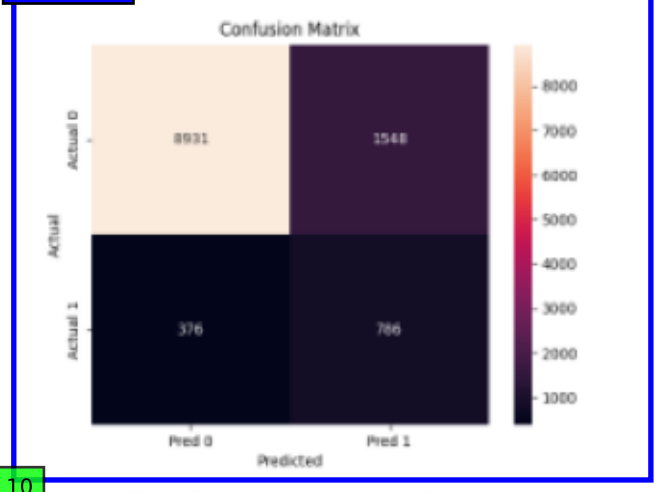


Figure 5: Confusion matrix for random forest model

3.6 ROC Curve and AUC

Image 3

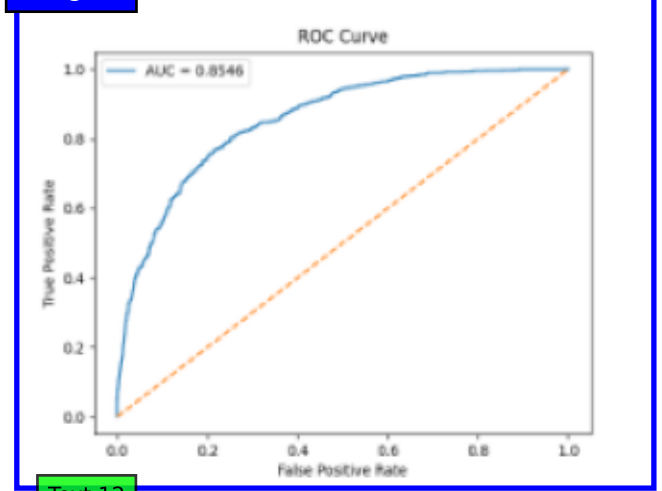


Figure 6: ROC CURVE and AUC

The Receiver Operating Characteristic (ROC) curve was used to evaluate the classification performance of the Random Forest model across different threshold values.

The Area Under the Curve (AUC) score obtained was:

$$AUC Score = 0.8546$$

Interpretation:

- The model shows strong separability between fraudulent and legitimate transaction classes.

- An AUC score greater than 0.85 indicates good classification performance.
- The ROC curve remains well above the diagonal reference line, demonstrating performance significantly better than random guessing.

3.7 Feature Importance Analysis

Feature importance scores obtained from the Random Forest classifier help identify which graph-based heuristic features contributed most to classification performance.

The top contributing features include:

- txId (or identifier-based feature)
- PageRank
- DFS Flow Score
- Depth
- In-degree

Insights:

- PageRank plays a major role, confirming the importance of graph centrality in fraud detection.
- Flow-based features such as DFS Flow Score effectively capture cyclic transaction behavior.
- Depth and degree features indicate transaction positioning and connectivity within the transaction network.
- Structural graph features are clearly informative and discriminative for fraud classification.

3.8 Key Observations

Based on the experimental evaluation of the Random Forest classifier trained on graph heuristic features, the following observations were made:

- Graph heuristics alone will not provide strong predictive power for fraud detection.
- The model achieves a moderate balance between:
  - Detecting fraudulent transactions (recall)
  - Avoiding false alarms (precision)
- Structural features such as PageRank, degree-based metrics, and flow scores are highly relevant indicators of suspicious transaction behavior.

3.9 Limitations

Despite strong overall performance, the Random Forest classifier exhibits certain limitations:

- Fraud class precision remains relatively low: 0.34
- The model still produces some false positive predictions.
- Classification performance depends on probability threshold tuning, where a threshold value of: 0.6 was used instead of the default 0.5 to improve prediction reliability.

4 Model Training and Evaluation — XGBoost on Graph Heuristics

4.1 Model Overview

To further improve performance on imbalanced fraud detection, an XGBoost classifier was trained using the same graph heuristic features.

XGBoost was selected for the following reasons:

- It handles class imbalance effectively via weighting
- It captures complex feature interactions through boosting
- It generally achieves higher predictive performance than bagging-based models

4.2 Data Preparation

Dataset used:

`graph_heuristics_features.csv`

Preprocessing steps:

- Unlabeled transactions (label = −1) were removed
- Feature set included 13 engineered graph heuristic features
- Target variable:
  - 0 → Legitimate transaction
  - 1 → Fraudulent transaction

Train-test split:

- Training set: 75%
- Test set: 25%

4.3 Handling Class Imbalance

Fraud detection datasets are highly imbalanced. To address this issue, the following weighting strategy was applied:

$$scale_pos_weight = \frac{No.of negative samples}{No.of positive samples}$$

This ensures the model assigns higher importance to fraudulent transactions during training.

4.4 Model Configuration

The XGBoost classifier was trained using the following hyperparameters:

- Number of estimators: 300
- Learning rate: 0.05
- Maximum depth: 6
- Tree method: `hist` (optimized for computational speed)
- Class weighting: `scale_pos_weight` (dynamic)
- Random state: 42

4.5 Evaluation Metrics

We trained and tested with random forests at different depths and splits and got the best model with an accuracy of 82.04%.

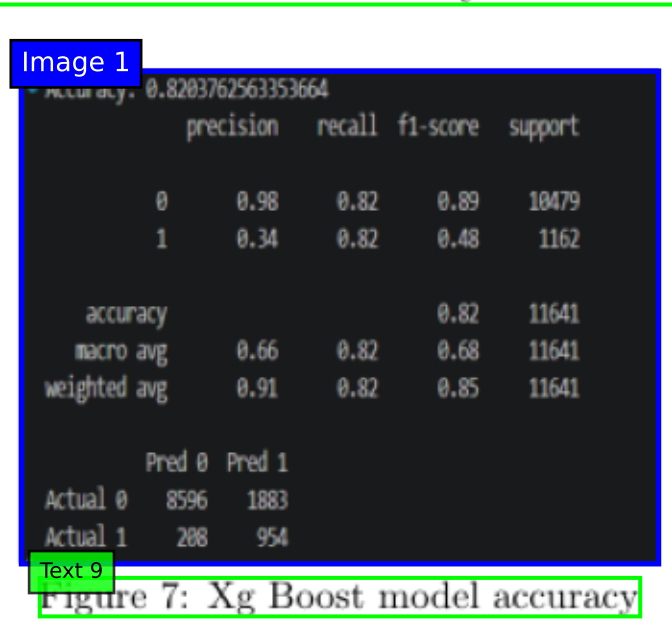


Figure 7: Xg Boost model accuracy

Key Insights:

- Fraud recall increased to 0.82, indicating strong improvement in fraud detection capability
- Precision for fraud remains relatively low, resulting in more false positive alerts
- The model prioritizes detecting fraud over avoiding false alarms

4.6 Confusion Matrix

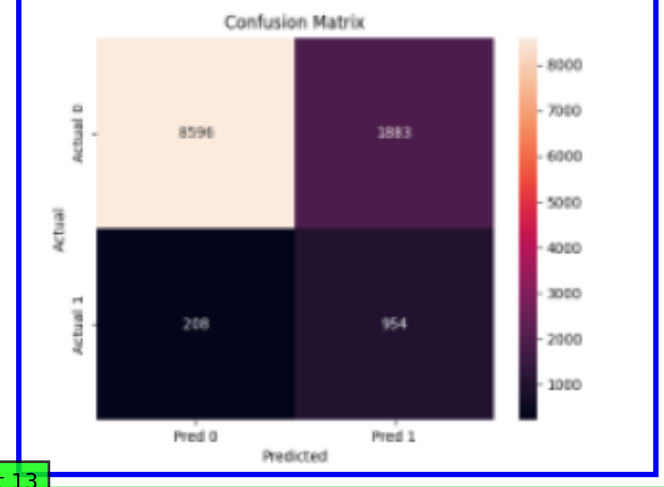


Figure 8: Confusion matrix for XGboost model

Interpretation:

- True Positives (TP): 954 (high fraud detection)
- False Negatives (FN): 208 (reduced missed fraud cases)
- False Positives (FP): 1883 (increased alert generation)

Compared to Random Forest:

- Better fraud recall
- Higher number of false positives

4.7 ROC Curve and AUC

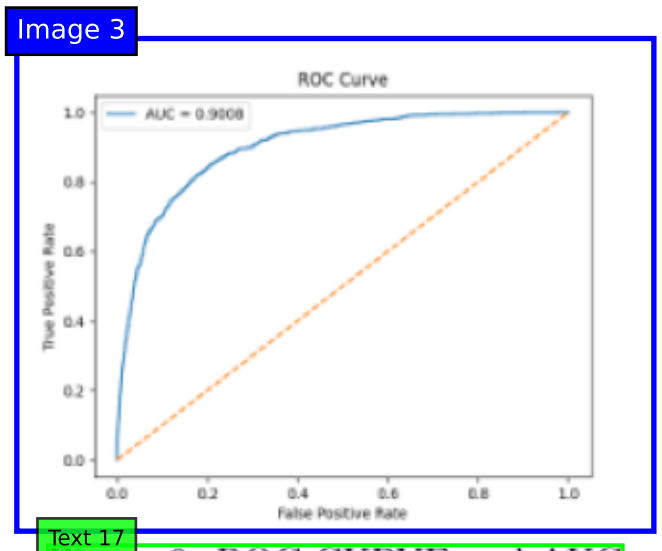


Figure 9: ROC CURVE and AUC

The ROC-AUC score obtained for the XGBoost classifier was:

$$AUC Score = 0.9008$$

Text 1

Interpretation:

- Strong improvement compared to Random Forest ($\sim 0.85 \rightarrow 0.90$)
- Indicates excellent separability between fraudulent and legitimate transactions
- Demonstrates strong ranking capability of the classifier

Text 2

4.8 Feature Importance Analysis

Text 3

Top contributing features include:

Text 4

- In-degree
- Out-degree
- DFS Flow Score
- Depth
- PageRank

Insights:

- Degree-based features dominate, confirming the importance of transaction connectivity
- Flow and depth features effectively capture transaction pathways
- PageRank contributes significantly, validating centrality-based fraud signals
- Structural graph features consistently drive model performance

Text 5

4.9 Comparison with Random Forest

Text 6

Table 2: Comparison Between Random Forest and XGBoost Models

Table 1

Metric	Random Forest	XGBoost
Accuracy	0.83	0.82
Fraud Recall	0.68	0.82
Fraud Precision	0.34	0.34
AUC	0.85	0.90

Text 7

- XGBoost significantly improves fraud detection recall
- The model achieves better ranking ability as reflected by the improved AUC score
- Slight reduction in accuracy occurs due to more aggressive fraud detection behavior

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5 Graph Neural Network (GCN) Model

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5.1 Model Motivation

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While heuristic features capture structural properties explicitly, they do not fully exploit relational dependencies between transactions. A Graph Neural Network (GNN) addresses this limitation by learning node representations through message passing over the graph structure.

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In this project, a Graph Convolutional Network (GCN) is used to combine:

Text 12

- Node features (original transaction features + heuristic features)
- Graph connectivity information (transaction flows)

Text 13

5.2 Graph Construction

Text 14

The transaction network is modeled as:

Text 15

$$G = (V, E)$$

Text 16

- Nodes (V): Transactions
- Edges (E): Directed fund transfers between transactions

Graph preparation steps:

- Features and heuristic datasets were merged using transaction identifiers (`txId`)
- Unlabeled nodes were removed from the dataset
- A mapping from transaction IDs to node indices was created
- Edges were converted into `edge_index` format for compatibility with PyTorch Geometric

Text 17

5.3 Feature Representation

Text 18

Each node representation consists of:

Text 19

- Original transaction features from the Elliptic dataset
- Graph heuristic features (degree, PageRank, SCC-based features, depth, motif scores, etc.)

This creates a hybrid feature space combining:

Text 20

- Raw attribute-level signals
- Structural graph-based signals

Text 1
5.4 Model Architecture

Text 2
The GCN architecture used in this project consists of three layers:

Text 3
- Layer 1:

Text 4
$GCNConv(input \rightarrow 128) + ReLU$

Text 5
- Layer 2:

Text 6
$GCNConv(128 \rightarrow 64) + ReLU$

Text 7
- Output Layer:

Text 8
$Linear(64 \rightarrow 2)$

Text 9
This architecture enables:

- Aggregation of neighborhood information
- Learning hierarchical node representations
- Capturing relational transaction behavior across multiple hops

Text 10
5.5 Training Strategy

Text 11
The GCN model was trained using the following configuration:

Text 12
- Optimizer:

Text 13
$Adam \quad (learningrate = 0.01)$

Text 14
- Loss function:

Text 15
$CrossEntropyLosswithclassweights$

Text 16
- Class weights:

Text 17
$[1.0, 2.0]$

Text 18
Higher penalty was assigned to fraud misclassification.

- Number of epochs:

Text 19
60

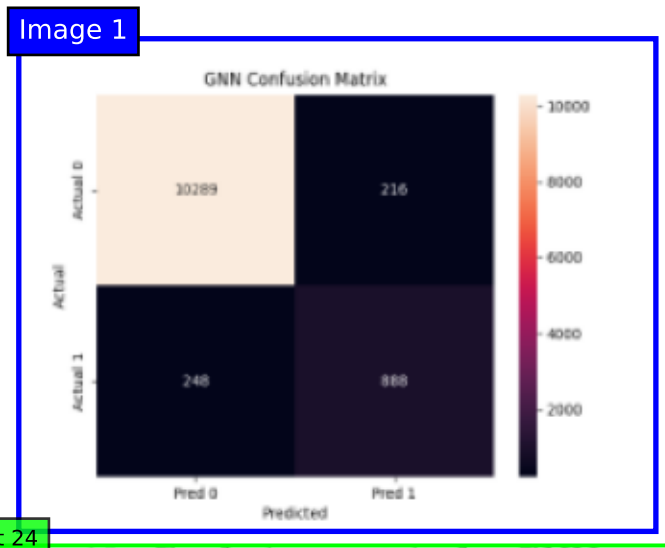
Text 20
- Train-test split:

Text 21
$75\% - 25\% \quad (stratified)$

Text 22
This configuration ensures:

- Better handling of class imbalance
- Stable convergence during training

Text 23
5.6 Confusion Matrix

Image 1


Text 24
Figure 10: Confusion matrix for GNN model

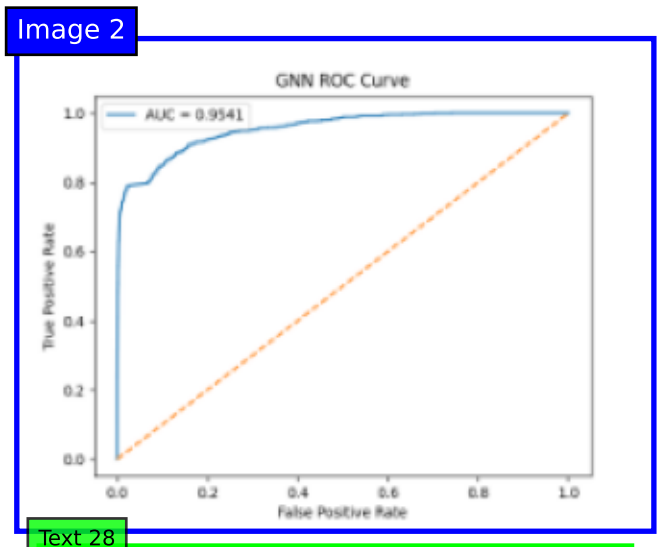
Text 25
Interpretation:

Text 26
- True Positives (TP): 888
- False Negatives (FN): 248
- False Positives (FP): 216
- True Negatives (TN): 10289

Key Insight:

- Very low false positives indicate high precision
- Strong fraud detection capability with balanced classification performance

Text 27
5.7 ROC Curve and AUC

Image 2


Text 28
Figure 11: ROC CURVE and AUC

Text 29
The ROC-AUC score obtained for the GCN classifier was:

Text 30
$AUCScore = 0.9541$

Text 31
Interpretation:

- Excellent separability between fraudulent and legitimate transaction classes

Text 1
- Significant improvement compared to:

Text 2
  - Random Forest (~ 0.85)
  - XGBoost (~ 0.90)

Text 3
5.8 Performance Analysis

Comparison of model performance:

Text 4
Table 3: Performance Comparison Across Models

Table 1

Metric	RF	XGB	GCN
Accuracy	0.83	0.82	0.96+
Fraud Recall	0.68	0.82	0.78
Fraud Precision	0.34	0.34	0.80+
AUC	0.85	0.90	0.95

Text 5
Observations:
- The GCN model achieves the best overall performance among all evaluated models
- It drastically reduces false positive predictions
- Strong fraud detection capability is maintained alongside improved classification precision

Text 6
6 Hybrid Model — GCN + XGBoost Ensemble

Text 7
Both models developed in this project capture complementary aspects of fraud detection:

Text 8
- **GCN:** Learns deep relational patterns from graph structure
- **XGBoost:** Uses explicit structural heuristics engineered from classical graph algorithms

Individually:
- GCN $\rightarrow$ High precision and strong structural learning
- XGBoost $\rightarrow$ High recall and strong minority-class detection

To leverage the strengths of both approaches, a weighted ensemble model was constructed.

Text 9
6.1 Ensemble Strategy

For each transaction node:

Text 10
$P_{GCN} = ProbabilitypredictedbyGCN$

Text 11
$P_{XGB} = ProbabilitypredictedbyXGBoost$

Text 12
The final ensemble probability is computed as:

Text 13
$$P_{final} = \alpha \cdot P_{GCN} + (1 - \alpha) \cdot P_{XGB}$$

Text 14
where:

Text 15
- α controls the contribution of the GCN model
- $(1 - \alpha)$ controls the contribution of the XGBoost model

Text 16
Final prediction rule:

Text 17
$$y = \{1\ if P_{final} > 0.50 otherwise$$

Text 18
6.2 Experimental Setup

Text 19
Prediction probabilities were obtained from:

Text 20
- Trained GCN model (`gnn_model.pth`)
- Trained XGBoost model (`xgboost_model.json`)

Text 21
The ensemble was evaluated across multiple values of:

Text 22
$\alpha \in \{0.0, 0.1, 0.2, \ldots, 1.0\}$

Text 23
Evaluation metrics:

Text 24
- Accuracy
- Precision
- Recall
- F1-score

Text 1

6.3 Results Across Different α and β Values

Image 1

	alpha_gcn	beta_xgb	accuracy	precision	recall	f1_score
0	0.0	1.0	0.832016	0.354498	0.878328	0.505125
1	0.1	0.9	0.853707	0.390026	0.884488	0.541341
2	0.2	0.8	0.884503	0.453081	0.884928	0.599315
3	0.3	0.7	0.916803	0.545516	0.884708	0.674891
4	0.4	0.6	0.950885	0.697447	0.877448	0.777161
5	0.5	0.5	0.966154	0.808692	0.855666	0.831516
6	0.6	0.4	0.966670	0.824691	0.836304	0.830457
7	0.7	0.3	0.966240	0.826345	0.828163	0.827253
8	0.8	0.2	0.965531	0.825509	0.820242	0.822867
9	0.9	0.1	0.964651	0.823621	0.811661	0.817598
10	1.0	0.0	0.963534	0.819529	0.803300	0.811333

Text 2

Figure 12: Results Across Different α and β Values

Text 3

6.4 Key Observations

Text 4

$\alpha = 0$ **(pure XGBoost):**

Text 5

- Very high recall (detects most fraud cases)
- Lower precision (more false positives)

$\alpha = 1$ **(pure GCN):**

- High precision
- Slightly lower recall

Balanced region ($\alpha \approx 0.5$–0.6)**:**

- Best overall performance
- Strong balance between precision and recall
- Reduced false alarms while maintaining strong fraud detection capability

Text 6

6.5 Best Model

The optimal configuration was obtained at:

Text 7

$\alpha = 0.5$

Text 8

Model performance:

- Accuracy: 96.6%
- Precision: 80.9%
- Recall: 85.6%
- F1-score: 83.2% (highest)

This represents a significant improvement compared to individual models.

Text 9

6.6 Interpretation

The ensemble model improves performance because:

Text 10

- GCN captures global graph structure

Text 11

- XGBoost captures explicit structural heuristics
- Their prediction errors are complementary rather than overlapping

This leads to:

- Reduced false positives
- Reduced false negatives
- Strong overall generalization performance

Text 12

6.7 Conclusion

The hybrid model demonstrates that combining graph neural networks with classical graph algorithm-based heuristic features yields superior fraud detection performance.

Text 13

Key takeaways:

- Graph structure plays a critical role in fraud detection
- Heuristic graph features provide strong predictive signals
- Graph Neural Networks capture deeper relational dependencies
- Ensemble learning effectively integrates both approaches for improved performance

Text 14

7 Final Model Evaluation — Hybrid Ensemble ($\alpha = 0.5$)

7.1 Model Configuration

The final model uses a balanced ensemble of:

Text 15

- GCN (50%) → captures relational graph patterns
- XGBoost (50%) → captures explicit structural heuristics

Final probability computation:

Text 16

$$P_{final} = 0.5 \cdot P_{GCN} + 0.5 \cdot P_{XGB}$$

Text 17

Prediction threshold used:

Text 18

0.5

Text 19

7.2 Overall Performance Metrics

Model performance:

Text 20

- Accuracy: 0.966
- Precision: 0.809

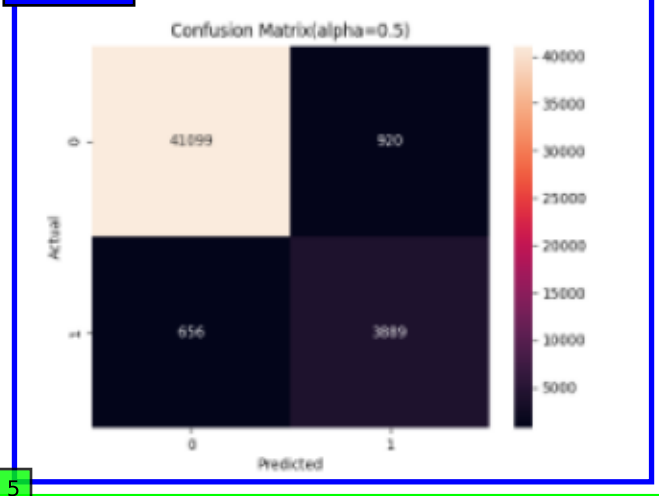
Text 1
- Recall: 0.856
- F1-score: 0.832

Text 2
Interpretation:

Text 3
- High accuracy with strong fraud detection capability
- Balanced precision–recall performance suitable for real-world deployment
- High F1-score indicates an optimal trade-off between false positives and false negatives

Text 4
7.3 Confusion Matrix Analysis

Image 1



Text 5
Figure 13: Confusion matrix for GNN + XGboost model(for α and β Values

Text 6
Breakdown:

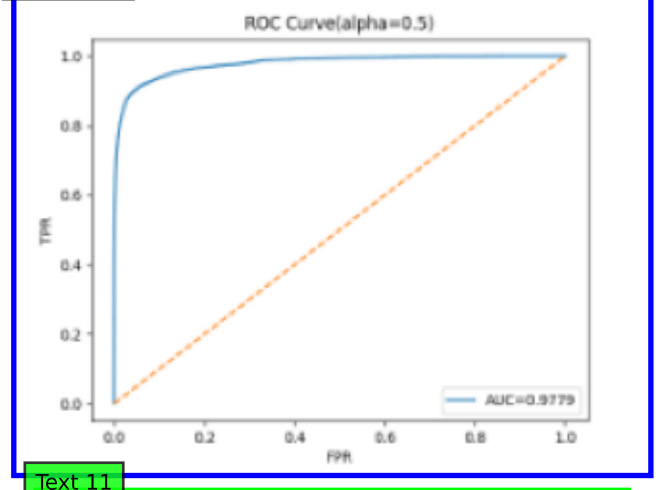
Text 7
- True Negatives (TN): 41099
- False Positives (FP): 920
- False Negatives (FN): 656
- True Positives (TP): 3889

Text 8
Insights:

Text 9
- Very low false positives minimize incorrect fraud alerts
- Strong number of true positives indicates effective fraud detection
- Balanced error distribution across prediction classes

Text 10
7.4 ROC Curve and AUC

Image 2



Text 11
Figure 14: ROC CURVE and AUC

Text 12
The ROC-AUC score obtained for the hybrid ensemble model was:

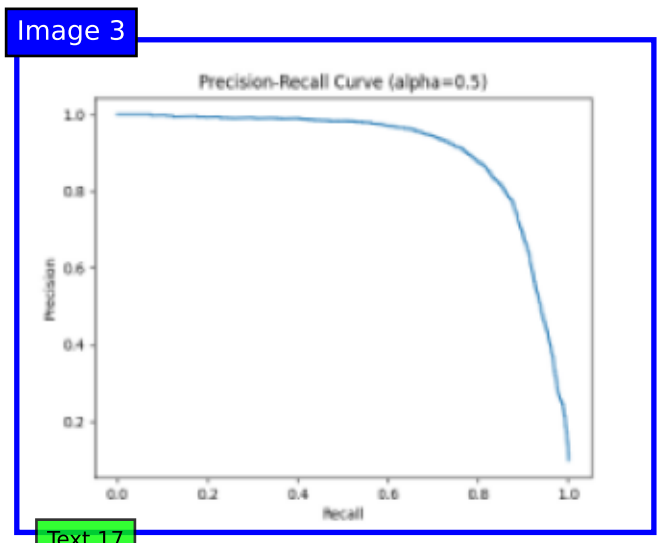
Text 13
$$AUC Score = 0.9779$$

Text 14
Interpretation:
- Near-perfect separability between fraudulent and legitimate transactions
- Significant improvement compared to:

Text 15
  - Random Forest (~ 0.85)
  - XGBoost (~ 0.90)
  - GCN (~ 0.95)

Text 16
7.5 Precision–Recall Curve

Image 3



Text 17
Figure 15: Precision–Recall Curve

Text 18
Precision–Recall (PR) analysis provides a more informative evaluation for imbalanced fraud detection datasets.

Text 19
Insights:
- Precision remains high across a wide range of recall values

Text 1
- Gradual decline in precision indicates stable model behavior under different thresholds
- Strong PR performance confirms robustness on highly imbalanced datasets

Text 2
7.6 Key Performance Insights

The hybrid ensemble significantly:

Text 3
- Reduces false positives compared to the XGBoost model
- Improves recall compared to the standalone GCN model
- Achieves the best F1-score across all evaluated models
- Demonstrates strong generalization performance on the full dataset

Text 4
7.7 Final Comparison

Text 5
Table 4: Final Performance Comparison Across Models

Table 1

Model	Accuracy	Precision	Recall	F1-score	AUC
Random Forest	0.83	0.34	0.68	0.45	0.85
XGBoost	0.82	0.34	0.82	0.48	0.90
GCN	~ 0.96	~ 0.80	~ 0.78	~ 0.79	0.95
Ensemble	0.966	0.809	0.856	0.832	0.978

Text 6
7.8 Final Conclusion

The final hybrid ensemble model demonstrates that combining graph-based deep learning with classical graph heuristics produces state-of-the-art fraud detection performance.

Text 7
Key conclusions:
- Graph structure is critical for identifying fraudulent behavior
- Heuristic graph features provide strong interpretable signals
- Graph Neural Networks capture deeper relational dependencies
- Ensemble learning effectively integrates both approaches for superior predictive performance